DISCUSSIONS

Paleomagnetism of Siluro-Devonian rocks from eastern Maine:[1] Discussion[2]

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Rock sequences in orogenic belts usually carry multi-component remanences composed of the initial (primary A′) remanence acquired when the rock was formed and secondary remanences (overprints B) produced during subsequent orogenic eposides. In some instances, remagnetization may have progressed to the point where the initial (A′) remanence has been erased completely (e.g., by heating, chemical, or hydrothermal alteration) and replaced by a remanence of orogenic age (B′). An initial remanence (A″) of orogenic age may also be carried by rocks newly formed during the orogeny. When more than one orogeny has affected the region, the process can be repeated, resulting in a mixture of initial (A′ or A″), first overprint (orogenic B′), second overprint (orogenic B″), ..., remanences. Hence the rocks can carry all or part of the signatures of the paleofield not only at the time that the rock was formed but also at the times of the subsequent orogenies. Thus, provided that the sequence of remanence acquisition can be identified, results from fold belts, although complicated, can provide much information about the displacement and integrity of orogenic terranes.

Because of the relation ½(tangent of inclination) = tangent of latitude, the paleolatitudinal displacement of the terrane between say time A_1' and time B_1' is easily determined by calculating the paleolatitudinal difference between pole A_1' and pole B_1' derived from remanences A_1' and B_1' observed in rock unit 1 (assuming that no tilting took place in the meantime). The occurrence or absence of tilt and more detailed mapping of the paleolatitudinal travel of the terrane can be defined by obtaining poles A_n', A_n'', B_n', B_n'', ..., from n rock units. However, all those poles should not be expected to fall on the apparent polar wander path (APWP) followed by the terrane because of local rotations about vertical axes that are common in orogenic belts (Irving 1964). Instead the observed poles will be deflected to the right or the left of the true APWP inasmuch as the rotations are dextral or sinistral.

It is obvious that before the provenance, the kinematics of travel, and the tectonic evolution of any terrane within fold belts can be mapped paleomagnetically it is necessary to sort out the initial from the orogenic remanences. If the remanences are misidentified, this will lead to incorrect tectonic interpretations. The purpose of this discussion is to argue that Kent and Opdyke in their study of the Eastport and Hersey Formations have misidentified an orogenic remanence as an initial one, and this has led them to incorrect conclusions in their evaluation of Siluro-Devonian data.

Throughout their paper, Kent and Opdyke are trying to promote the idea that the observed remanence was acquired prior to tilting and is therefore representative of the field direction at the location at about the time that the rocks were formed (A′ remanence). However, they do not provide any evidence for a pre-tilting remanence acquisition. They invoke baked contact relationships and the occurrence of dual polarities; these arguments have no direct relevance to the pre-tilting hypothesis. Their major argument is that "the similarity in directions over different rock types and magnetic carriers point to an acquisition of magnetization near to the time of rock formation" (Abstract). However, this argument is not supported by the data, which, apparently, they have neglected to analyse in detail. Hence, by averaging the results irrespective of rock types (their Tables 1 and 2), they have not shown but have rather accepted as a basic assumption that all rock types yield similar directions. When dealing with different rock types, one approach is to compute the mean directions obtained from each rock type separately and to test by the statistical methods available whether the mean directions are similar or not. The results of such tests performed on the data sets (before and after tilt correction) reported by Kent and Opdyke show that the coherence among the mean directions of the individual rock types greatly deteriorates when correction for tilt is applied, thus invalidating the pre-tilting hypothesis. The reanalysis shows that a similarity in directions does exist but it is found among mean directions obtained *without correcting for tilt*, thus indicating a post-tilting magnetization for, at least, five of the rock types sampled.

[1]Paper by D. V. Kent and N. D. Opdyke. 1980. Canadian Journal of Earth Sciences, **17**, pp. 1653–1665.

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Results from the Hersey and Eastport Formations are compiled in Table 1, calculated from Tables 1 and 2 of Kent and Opdyke. There are six groups arranged by rock type, except for group 3 where the sill sampled from the Eastport Formation (one site) has been included with the seven lava sites, following Kent and Opdyke. Mixed polarities were encountered in two groups (2M and 3M), and the results of the normal (N) and reverse (R) polarities have also been computed separately. Inspection shows that the two polarity directions are not antiparallel, with divergencies from antiparallelism ranging from 17 to 35° of arc. Evidently, either the N and R polarity remanences were not acquired contemporaneously or they have not been successfully separated. In either case, the averaging of the directions (2M and 3M) irrespective of polarity most likely introduces a direction bias and the 2R and 3R results probably provide one with better *estimates* of the field direction at the time of acquisition of the reverse polarity remanence than the 2M and 3M results. This is also indicated by the fact that the precision (k) of the R mean direction is, *in all instances*, greater than the k of the corresponding M mean direction, up to 2.7 times larger. It would then appear that the normal polarity directions are a source of error for estimating the direction of the reverse polarity field, and their inclusion in the averaging process, as done by Kent and Opdyke, is questionable. Therefore, in the following analysis, only the reverse polarity mean directions (groups with a suffix R) are considered.

A widely used test for estimating if the remanence of a particular rock unit was acquired before or after folding (or tilting) is the comparison of k ratios (Watson 1956) before (k_1) and after (k_2) correction for tilting. A ratio of $k_2/k_1 > 1$ indicates that the dispersion of directions is less after than before correction, thereby favouring a pre-folding remanence acquisition time; a ratio of $k_1/k_2 > 1$ indicates that the dispersion increases when correcting for tilt, thereby favouring a post-folding remanence acquisition time. Only three groups (1R, 2R, 3R) have a sufficient number of sites for the application of this statistical test and the results (Table 1) favour a pre-folding remanence for the Hersey sills (1R) and a post-folding remanence for both the Hersey red beds (2R) and the Eastport lavas and sill (3R). However, none of the ratios is large enough to be statistically significant at the 95% level of confidence (McElhinny 1964) so that this test alone is not conclusive without supporting evidence.

Additional evidence that not only the 2R and 3R but also the 4R, 5R, and 6R groups were all magnetized after tilting is provided by comparing all possible pairs of group mean directions, both before and after tilt correction. The results presented in Table 2 are most revealing.

Firstly, comparisons of angular differences among the four units (3R, 4R, 5R, 6R) of the Eastport Formation show convincingly that the remanence observed was acquired after tilting rather than before tilting; in all possible combinations, the differences increase markedly (up to 19 times) when correction for tilt is applied. There can be little doubt that the age of the Eastport remanence is post-tilting. Likewise, when the comparison is made between the 2R unit and each of the 3R to 6R units, indications are that the age of magnetization of the Hersey red beds is also post-tilting. Aside from the fact that correction for tilt *always* increases the angular difference, it is important to note that, in all 10 combinations (2R to 6R), the "before tilt correction" differences are reasonably small. The five units were therefore probably all magnetized in the same field. This conclusion is reinforced by the F-distribution tests (Table 2) showing that, *without correction for tilt* (F_1), those five groups are not significantly different and therefore belong to the same population.

Secondly, the F-distribution tests of 1R with 2R and 3R are significant, indicating that 1R belongs to a different population. However, although the tests show that 1R does not have the same remanence direction as the other rock units, it cannot be established whether the magnetization occurred before or after tilting. The F_2/F_1 ratio (Table 2) tends to favour a post-tilting time. Conversely, the k_2/k_1 ratio (Table 1) tends to favour a pre-tilting time. The results for the angular differences (Table 2) are inconclusive, the values before and after (tilt correction) being comparable. Hence, in this instance, none of the tests are conclusive and the remanence age of the Hersey sills remains unknown.

Thirdly, the F-distribution results contradict the argument of Kent and Opdyke (p. 1661) that

> similar characteristic directions ... obtained from a variety of rock types ... point to a magnetization age that is near to the age of formation of the rock units rather than by some later remagnetization. The formation mean directions corrected for bedding tilt may therefore be taken to approximate the paleomagnetic field direction for Siluro-Devonian time, ..."

On the contrary, the F-distribution tests show this argument to be false. All tests (F_2 in Table 2) are significant, showing that, if tilt correction is applied, each of the six groups belongs to different populations. Although the level of confidence of the last three tests is only 90%, the very large F_2/F_1 ratios leave little doubt about the incorrectness of the deductive statement quoted above.

A compilation of the results using different combinations of groups is given in Table 3. All combinations show unequivocally that the remanence was acquired after tilting; the *negative* fold tests are significant at the 99 and 99.5% level. A graphical representation of the

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TABLE 1. Mean directions and statistics obtained from Tables 1 and 2 of Kent and Opdyke for different rock types (groups). Groups yielding mixed polarities (M) have also been decomposed into reverse (R) and normal (N) polarity groups

		Before tilt correction					After tilt correction					Fold test	
Group	N	$D°, I°$	R	k_1	$\alpha_{95}°$	Long.°, Lat.°	$D°, I°$	R	k_2	$\alpha_{95}°$	Long.°, Lat.°	k_2/k_1	k_1/k_2
						Hersey Formation							
1R sills	7	186.4, +41.1	6.7723	26	12	287E, 21S	161.5, +50.7	6.8921	56	8	310E, 12S	2.2	
2M red beds	9	181.1, +26.0	8.8752	64	6	292E, 31S	164.8, +34.0	8.8573	56	7	309E, 25S		1.1
2R	8	180.7, +27.8	7.9217	89	6	293E, 30S	163.1, +35.1	7.8934	66	7	310E, 24S		1.3
2N	1	008.3, −11.0	—	—	—		356.9, −23.9	—	—	—			
						Eastport Formation							
3M lavas + sill	8	185.9, +14.8	7.5102	14	15	286E, 37S	180.4, +33.9	7.6461	20	13	293E, 27S	1.4	
3R	6	188.7, +23.1	5.8699	38	11	283E, 33S	180.6, +39.6	5.7915	24	14	292E, 23S		1.6
3N	2	357.6, +11.5	1.9269	—	—		359.9, −16.7	1.9716	—	—			
4R baked cont.	2	196.9, +17.8	1.9782	—	—	285E, 36S	149.0, +40.5	1.9778	—	—	323E, 16S		
5R red beds	2	191.1, +26.4	1.9182	—	—	280E, 30S	184.7, +49.5	1.9840	—	—	289E, 14S		
6R calc. nodules	2	186.4, +20.2	1.9965	—	—	285E, 34S	198.0, +34.5	1.9966	—	—	274E, 24S		

NOTES: In both groups 2 and 3, the two polarities are far from antiparallelism. The divergences between 2N and 2R are 18 and 17° of arc before and after tilt correction, respectively; between 3N and 3R, they are 35 and 22°. The 2N and 3N poles are not given owing to the uncertain significance of the normal directions (see text). N is number of sites; $D°$ and $I°$ are declination and inclination; R is resultant vector; k is precision estimate; α_{95} is radius of circle of confidence (Fisher 1953); k_2/k_1 is ratio for statistical significance of positive fold test (McElhinny 1964); k_1/k_2 is ratio for statistical significance of negative fold test (this article). Of the five tests shown in this table, none is significant at the 95% level of confidence.

TABLE 2. Comparison of group mean directions of different rock types given in Table 1

Groups	N', N''	Difference in degrees of arc: Before tilt correction	Difference in degrees of arc: After tilt correction
1R, 2R	7, 8	14	16
1R, 3R	7, 6	17	18
2R, 3R	8, 6	09	14
2R, 4R	8, 2	11	13
2R, 5R	8, 2	10	21
2R, 6R	8, 2	09	28
3R, 4R	6, 2	04	24
3R, 5R	6, 2	04	10
3R, 6R	6, 2	05	14
4R, 5R	2, 2	09	26
4R, 6R	2, 2	02	38
5R, 6R	2, 2	08	18

		F-distribution test*: Before tilt correction		After tilt correction		
Groups	N', N''	F_1	Significant	F_2	Significant	F_2/F_1
1R, 2R	7, 8	4.80	Yes at 95%	8.29	Yes at 99%	1.7
1R, 3R	7, 6	4.83	Yes at 95%	5.00	Yes at 95%	1.0
2R, 3R	8, 6	2.48	No at 95%	4.11	Yes at 95%	1.6
4R, 5R	2, 2	0.26	No at 90%	5.52	Yes at 90%	21.2
4R, 6R	2, 2	0.07	No at 90%	17.64	Yes at 90%	252.0
5R, 6R	2, 2	0.20	No at 90%	5.01	Yes at 90%	25.1

*Watson (1956). $F = (N' + N'' - 2)[R' + R'' - R_2/(N' + N'')]$ where N' and N'' are the numbers of sites of the two groups; R' and R'' are the resultant vectors of each group; and R_2 is the resultant vector of the two groups combined. F_2/F_1 provides one with a comparison of intermixing of directions before and after tilt correction with a ratio >1 indicating less scatter before than after applying the correction. The larger the ratio, the greater the probability that the remanence was acquired after tilting.

poles is shown in Fig. 1 together with the means and ovals of confidence obtained when combining the Eastport Formation rock units with the Hersey red beds. If the Hersey red beds are excluded, the configuration remains essentially the same except that the axes of the oval of confidence *after* tilt correction increase by about 30% (Table 3).

In summary, the analysis indicates that, except possibly for the Hersey sills, all rock units were magnetized essentially contemporaneously but some time after tilting. The post-tilting direction of 187°, +23° yields a pole (EH) at 285°E, 33°S, a considerable distance away from the (pre-tilt) poles quoted by Kent and Opdyke. It cannot be established whether the age of the Hersey sills remanence is pre- or post-tilting (subsequently denoted H_c and H_u, respectively). For directions and poles, see 1R in Table 1. Even though the exact ages of these poles cannot be established, probable age limits can be delineated. The fact that the EH remanence is post-folding rules out a Silurian age. It is unlikely to be younger than Carboniferous because of its direction. The age therefore is confined to the Devonian–Carboniferous interval. Since the same remanence direction was recorded in five different rock types, it was probably produced (as a B′ type remanence) during an orogenic event, the Acadian or the Alleghenian. The Hersey sills yield a different direction. The remanence could be initial (A′ remanence = H_c), having survived or escaped the orogenic overprint that produced the EH remanence, or it could be the signature of an orogenic event (B′ remanence = H_u) other than the orogenic event that imprinted the EH remanence. Even considering the lack of precision in remanence ages, these results are, nonetheless, an important contribution towards the

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FIG. 1. Poles derived from each of the six rock types sampled by Kent and Opdyke. The numbers correspond to the group numbers (Table 1 and text). Poles are identified according to the formation and to whether correction for tilt has been applied (triangles) or not (circles). Poles obtained by averaging the results of groups 2R, 3R, 4R, 5R, and 6R before subsequently denoted pole EH) and after applying tilt correction are shown with their ovals of confidence (P at 0.05). Results of other combinations indicating a post-tilting remanence acquisition are given in Table 3. SA is the sampling area.

eventual unravelling of the Appalachian tectonic history. This is shown by incorporating these results into the presently available data base.

Figure 2 shows the poles derived from Ordovician to Devonian rock units located in the eastern Appalachians (horizontal hatching). None of them can be unambiguously classified as a pole derived from an initial remanence. However, some of them are clearly derived from overprints probably produced during orogenic events, e.g., poles (EH, D) derived from post-folding remanences. In those instances where multi-component remanences have been observed, only one of the poles (e.g., 80 and 81; CB1 to CB4) can be an initial one, the others being overprints. The wide scatter of poles (Fig. 2) testifies to the complexity of paleomagnetic results from fold belts. The question is: can a generally coherent hypothesis be drawn from such widely scattered results? As a first approach to the problem, I wish to present what can be called the simplest possible polar path (SPPP) and show that it is consistent with the data.

The paleolatitudinal distribution (inset, Fig. 2) shows that the hatched (or part of the) area moved latitudinally by some 60° in the Ordovician to Carboniferous interval. I suggest that the peak observed in the 10–20° interval

TABLE 3. Analysis of the grand mean directions and statistics of the different rock types of the Eastport and Hersey Formations*

	Before tilt correction							After tilt correction							
						Pole					Pole			Negative fold test	
Groups	N	$D°, I°$	R	k_1	$\alpha_{95}°$	Long.°, Lat.°	$\delta p°, \delta m°$	$D°, I°$	R	k_2	$\alpha_{95}°$	Long.°, Lat.°	$\delta p, \delta m$	k_1/k_2	Significant
Eastport sediments															
4R+5R+6R	3	188, 21	2.9923	259	8	283E, 33S	5, 8	178, 43	2.8724	16	32	295E, 20S	26, 40	16.2	Yes at 99%
Eastport, all rock types															
3R+4R+5R+6R	4	188,22	3.9919	372	5	283E, 33S	3, 5	178, 42	3.8702	23	20	294E, 20S	16, 25	16.2	Yes at 99.5%
Eastport Formation and Hersey red beds															
2R+3R+4R+5R+6R	5	187, 23	4.9811	211	5	285E, 33S	3, 5	175, 41	4.8464	26	15	297E, 21S	12, 18	8.1	Yes at 99%

NOTE: δp and δm are the semi-axes of the ovals of confidence ($P = 0.05$).
*The Hersey sills are not included in the analysis because of the high probability that the mean direction is significantly different from the mean directions of other groups, both before and after tilt correction (Table 2).

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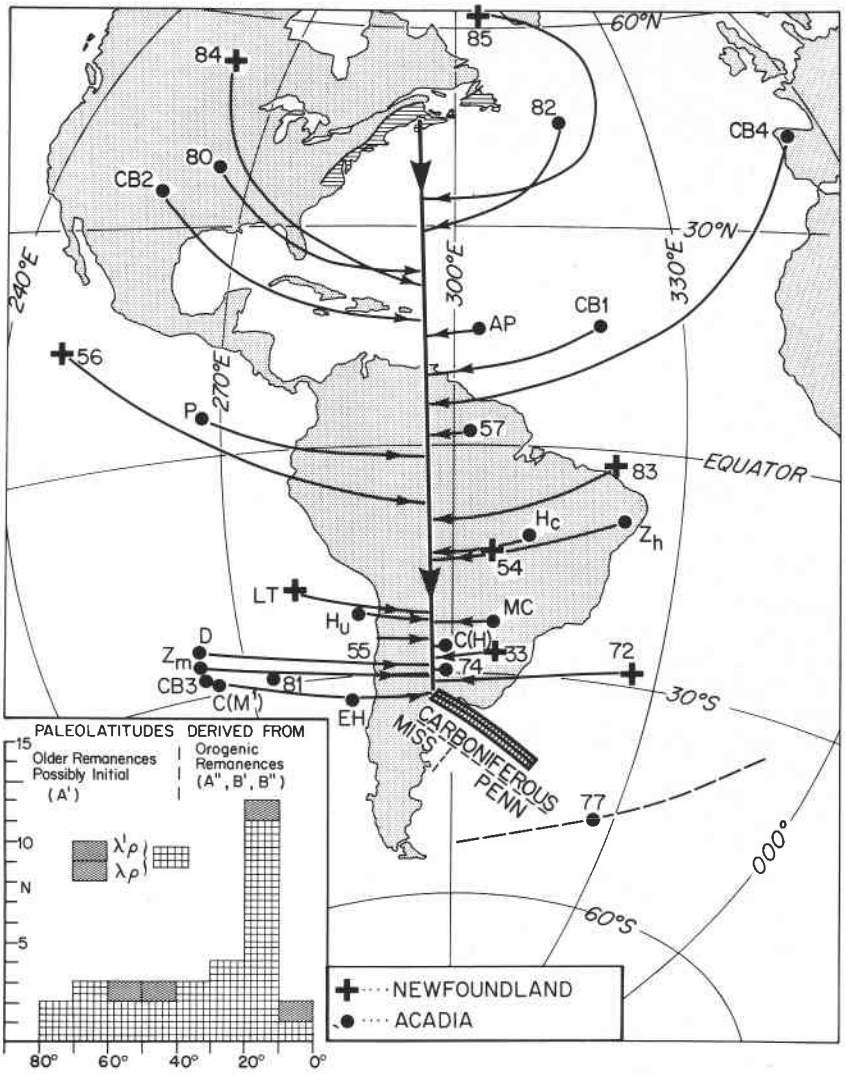


FIG. 2. Paleopoles (south) derived from Ordovician to Devonian rock units located within the hatched area shown. The numbered poles are keyed to the listing of Roy *et al.* (1979). Other poles are CB1 to CB4 (Rao *et al.* 1981); P, $C(M^1)$, C(H), Z_m, Z_h (Roy and Anderson 1981); AP (Rao and Van der Voo 1979); D (Brown and Kelly 1980); LT (Lapointe 1979); MC (Schutts *et al.* 1976); H_U and H_C are from the Hersey sills *in situ* and corrected for tilt, respectively (this article, Table 1); EH is the combined Eastport Formation and Hersey red beds pole (this article: third entry in Table 3 before tilt correction; see also caption of Fig. 1). Rotations about vertical axes to bring the individual poles into the SPPP are shown. This does not mean that the movement of the locality was as simple, e.g., tilting may have occurred. Hence, pole 77 cannot be brought into the SPPP, meaning that the pluton may have been tilted as suggested by Ashwal and Hargraves (1977) or its remanence is post-Carboniferous. Note: the age of the AP pole is associated with the age of metamorphism of the Arden pluton (Rao and Van der Voo 1979). Pole CB4 is calculated from the NE-3 direction found at 4 sites (Rao *et al.* 1981). The Carboniferous APWP segment (Late Mississippian to Permo-Carboniferous boundary) is from Roy and Morris (1981). The inset shows the paleolatitudinal distribution (in 10° intervals) of the 30 observations using two methods: λ_p calculated from the inclination of the observed remanence and λ_p^1 calculated with respect to the reference localities 293°E, 46°N for results from Acadia and 304°E, 49°N for results from Newfoundland. It is argued that the low paleolatitudes are, in general, derived from orogenic remanences of Devonian–Carboniferous ages whereas the intermediate to high paleolatitudes are derived from older remanences.

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(where the EH result falls) is most probably attributable to orogenic remanences acquired during both the Acadian and Alleghenian orogenies. The intermediate to high paleolatitudes result from older, possibly initial, remanences and provide one with a record of the provenance and movements of the sampling localities. The age sequence high paleolatitude → low paleolatitude is preferred because (1) the reverse sequence would necessitate much higher rates of (and more complicated) displacements; and (2) rock units that have yielded high paleolatitude results have also, in general, yielded low paleolatitude results that often agree with results obtained from post-folding remanences. The implication therefore is that only the high paleolatitude results are derived from pre-orogenic remanences.

The SPPP is shown in Fig. 2. Leaving aside interpretation of individual poles, it consists of a simple paleolatitudinal path passing roughly through the center of the observed poles. It commences at the reference locality for Acadia and terminates at the beginning (Late Mississippian) of the well defined Carboniferous APWP segment obtained from the east coast Carboniferous basin (Roy and Morris 1981). The divergence of individual poles from the SPPP is attributed to rotations of the sampling area about local vertical axes.

Does the overall apparent northward displacement (south poles are shown in Fig. 2) of some 60° correspond to a reasonable rate of displacement? The oldest poles are the AP (440 Ma, Rao and Van der Voo 1979) and CB1 (430–460 Ma, Rao *et al.* 1981). The youngest poles could be overprints produced during the Alleghenian orogeny (300–320 Ma), which affected at least part of the hatched area, the Maritime disturbance of Poole (1967). With these two limits, we find a very reasonable average paleolatitudinal displacement of 5 cm/year. The poles lying at the younger end of the SPPP (20–35°S) are good candidates for an Alleghenian overprint; that is, they may well be derived from remanences acquired during Middle to Late Carboniferous (Reynolds *et al.* 1981). Their lying slightly north of the Carboniferous segment may be owing to the bias of an older remanence, be it the initial one or an Acadian orogenic overprint. As for the older end of the SPPP, it cannot be established if the displacement followed a continuous or a back-and-forth motion. All the poles north of 15°S are believed to be in the 380–460 Ma interval. If the poles are ordered according to the ages given by the respective authors, a back-and-forth movement is required since poles 80, 82, 84, and 85 are given ages of 380–410 Ma and poles CB1, CB2, CB3, AP, 56, 57, and P are included in the 410–460 Ma interval. Such back-and-forth movement would imply 8–10 cm/year average paleolatitudinal displacement. However, considering both the errors associated with radiometric dating and the fact that these poles may not all be derived from initial remanences, the back-and-forth movement is far from being proven. Also to be considered is the possibility that the whole of the hatched area was not integral from its conception but is the result of separate parts each having its own travel path. This alternative would reduce the angle of rotation presently required with the single SPPP as shown in Fig. 2. Investigations of these possibilities must await the availability of further data. Nonetheless, in my view, there is now sufficient evidence with several poles that yield intermediate to high paleolatitudes and are derived from different localities to suggest that the hatched area (or its parts) originated a considerable distance away from its Carboniferous location and that the poles yielding paleolatitudes similar to those expected for the Carboniferous are recording orogenic events. The (post-tilting) pole EH derived from the data of Kent and Opdyke falls into this latter group. It is also of interest that the H_c pole derived from a possibly pre-tilt remanence lies much further to the north along the SPPP and therefore is consistent with the interpretation presented here.

It is noted that the present data do not provide evidence for or against dislocation between Acadia and Newfoundland, nor do they provide evidence for or against detachment of either area from the North American craton. The estimated rates of displacement are not too high for a reasonable rate of movement of an entire craton. Whether or not the northward displacement observed in the Appalachians is applicable to the craton as a whole remains to be seen. Further data from the craton are needed for a meaningful comparison.

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ASHWALL, L. D., and HARGRAVES, R. B. 1977. Paleomagnetic evidence for tectonic rotation of Belchertown pluton, west central Massachusetts. Journal of Geophysical Research, **82**, pp. 1315–1324.

BROWN, L. L., and KELLY, W. M. 1980. Paleomagnetic results from northern Maine — reinterpretations. Geophysical Research Letters, **7**, pp. 1109–1111.

FISHER, R. A. 1953. Dispersion on a sphere. Proceedings of the Royal Society of London, Series A, **217**, pp. 295–305.

IRVING, E. 1964. Paleomagnetism and its application to geological and geophysical problems. John Wiley and Sons, New York, NY. 399 p.

LAPOINTE, P. L. 1979. Paleomagnetism of the Notre Dame Bay lamphrophyre dikes, Newfoundland, and the opening of the North Atlantic Ocean. Canadian Journal of Earth Sciences, **16**, pp. 1823–1831.

McElhinny, M. W. 1964. Statistical significance of the fold test in paleomagnetism. Geophysical Journal of the Royal Astronomical Society, **8**, pp. 338–340.

Poole, W. H. 1967. Tectonic evolution of Appalachian region of Canada. *In* Geology of the Atlantic region. *Edited by* E. R. W. Neale and H. Williams. The Geological Association of Canada, Special Paper No. 4, pp. 9–52.

Rao, K. V., and Van der Voo, R. 1979. Paleomagnetism of a Paleozoic anorthosite from the Appalachian Piedmont, northern Delaware: possible tectonic implications. Earth and Planetary Science Letters, **47**, pp. 113–120.

Rao, K. V., Seguin, M. K., and Deutsch, E. R. 1981. Paleomagnetism of Siluro-Devonian and Cambrian granitic rocks from Avalon zone in Cape Breton Island, Nova Scotia. Canadian Journal of Earth Sciences, **18**, pp. 1187–1210.

Reynolds, P. H., Zentilli, M., and Muecke, G. K. 1981. K–Ar and $^{40}Ar/^{39}Ar$ geochronology of granitoid rocks from southern Nova Scotia: Its bearing on the geological evolution of the Meguma Zone of the Appalachians. Canadian Journal of Earth Sciences, **18**, pp. 386–394.

Roy, J. L., and Anderson, P. 1981. An investigation of the remanence characteristics of three sedimentary units of the Silurian Mascarene Group of New Brunswick, Canada. Journal of Geophysical Research, **86**, pp. 6351–6368.

Roy, J. L., and Morris, W. A. 1981. Paleomagnetic results from the Carboniferous of North America; the development of a chronostratigraphic marker horizon for Canadian east coast Carboniferous strata. Compte-Rendu, 9th International Congress of Carboniferous Stratigraphy and Geology. (In press.)

Roy, J. L., Anderson, P., and Lapointe, P. L. 1979. Paleomagnetic results from three rock units of New Brunswick and their bearing on the lower Paleozoic tectonics of North America. Canadian Journal of Earth Sciences, **16**, pp. 1210–1227.

Schutts, L. D., Brecher, A., Hurley, P. M., Montgomery, C. W., and Krueger, H. W. 1976. A case study of the time and nature of paleomagnetic resetting in a mafic complex in New England. Canadian Journal of Earth Sciences, **13**, pp. 898–907.

Watson, G. S. 1956. Analysis of dispersion on a sphere. Monthly Notices of the Royal Astronomical Society, Geophysical Supplement, **7**, pp. 153–159.

Paleomagnetism of Siluro-Devonian rocks from eastern Maine:[1] Reply[2]

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Introduction

Roy's discussion of our paper covers two topics. First, he attempts to show on the basis of various statistical tests that the characteristic magnetizations we reported in the Hersey red beds and the Eastport Formation were acquired after tectonic tilting and represent what he refers to as an "orogenic" remanence. Second, he presents an interpretation of paleomagnetic data from eastern Appalachian rock units proposing that low paleolatitudes are in general derived from orogenic remanences of Devonian–Carboniferous age whereas the intermediate to high paleolatitudes are derived from older remanences. In reply, we argue that Roy's statistical methods may not be appropriate and nevertheless are not sufficient to demonstrate a post-folding origin of remanence and that the weight of evidence in fact favors remanence acquisition near the time of formation of the Siluro-Devonian rock units. Independent of the question of pre- or post-folding magnetization of these rock units, we find that the available paleomagnetic data from the eastern Appalachians do not support his model of a relatively simple paleolatitudinal progression with age for this area, but rather point to the results from Acadian (Devonian) intrusive rocks as anomalous for reasons that are not yet entirely clear.

Age of magnetization

Instead of analyzing the dispersion among the site means according to formation as we did, Roy first groups them according to six rock types. He then calculates that the dispersion among five of the rock type

[1]Discussion by J. L. Roy. 1982. Canadian Journal of Earth Sciences, **19**, pp. 225–232.

[2]Lamont-Doherty Geological Observatory Contribution No. 3232.

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